

UNIT 4: Weather Analysis Products

d. Skew-T Analysis

Skew-T Introduction

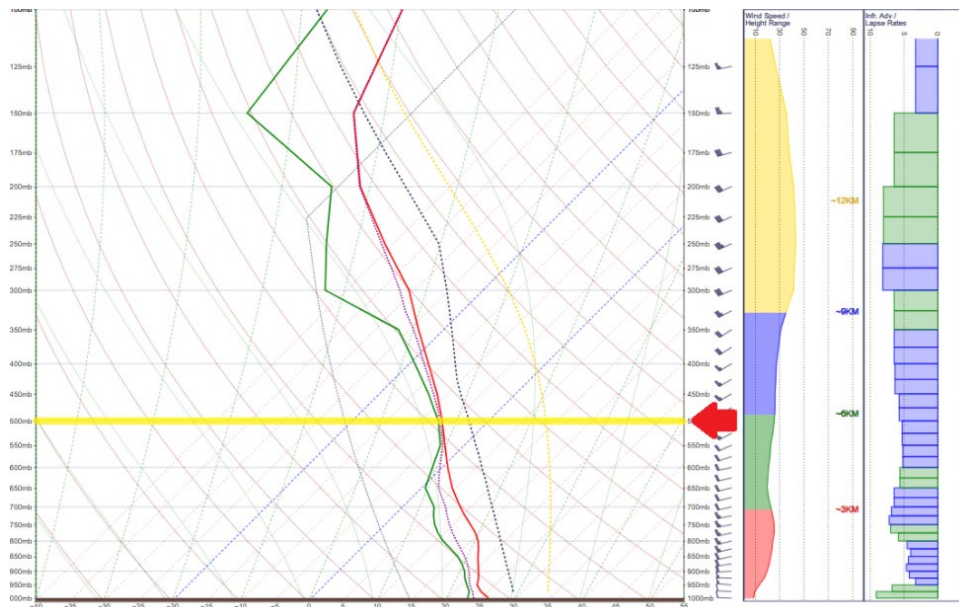
Skew-Ts are one of the most versatile tools for a weather forecaster. A Skew-T is a product that depicts temperature, dew point, wind, pressure, stability, and many other things as it relates to a specific location at a specific time. Many of the weather elements discussed in previous units can be analyzed on a Skew-T. They are extremely useful when analyzing severe weather parameters. Becoming familiar and competent with this product will greatly increase your effectiveness as a weather forecaster. Before learning how to analyze a Skew-T, we need to get familiar with the basic features of it.

The goal of this objective will be to show you have the capability to perform a basic Skew-T satellite analysis. To successfully demonstrate competency on this topic, you will be given information through lecture, a series of questions and tasks designed to display your ability to correctly identify weather data and significant indices discussed throughout this objective. These foundational analysis concepts will be measured with a Progress Check (PC) at the end of the objective and must be passed to continue onto the next objective. Follow your instructor's guidance as they provide information and instruction for completing all requirements necessary to pass this objective.

INPUT SKEW-T SOPS HERE

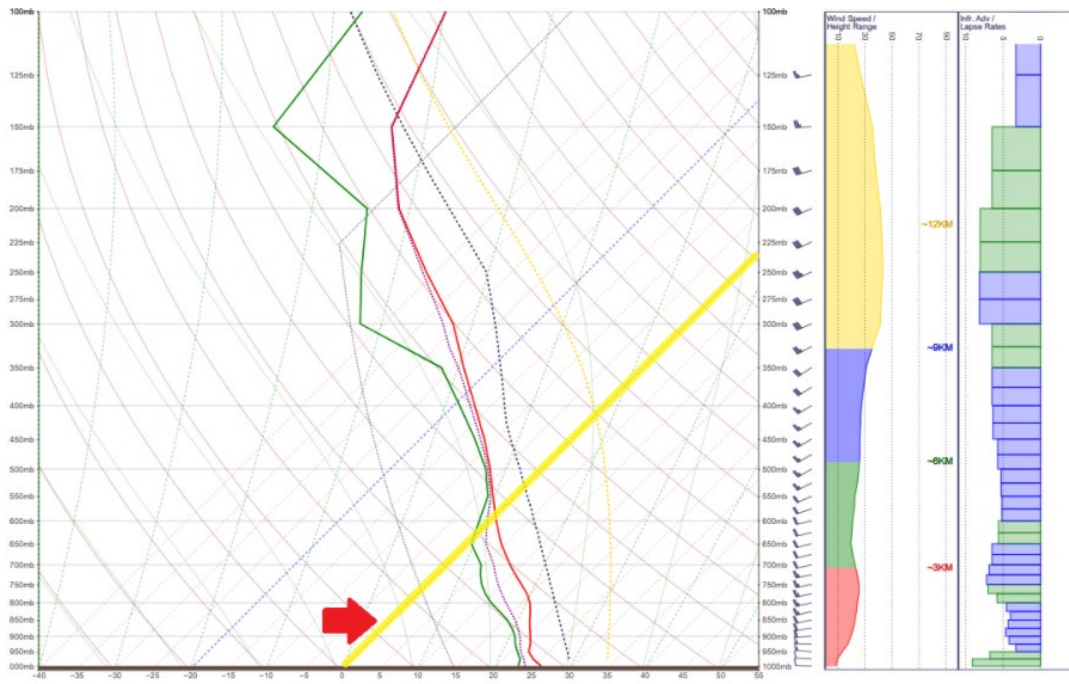
Isobars

Isobars are lines of equal pressure. They are horizontal brown lines drawn at 10 millibar (mb) intervals and labeled every 50 millibars. Note the spacing between isobars is close at the bottom and widens aloft. This is because the spacing reflects the expansion of our atmosphere as height increases.



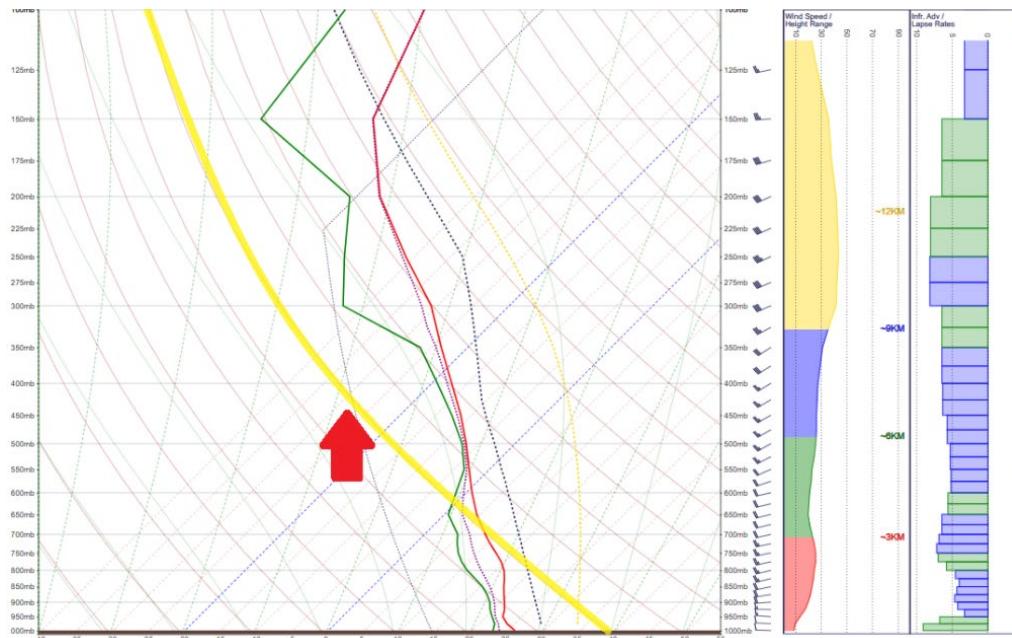
Isotherms

Isotherms are lines of equal temperature. They are diagonal brown lines extending from the lower left to the upper right (Figure 1-2, right side). Their interval is one degree Celsius with labels every five degrees. Also, there is a Fahrenheit scale along the bottom of the chart. You can use it to convert temperatures to Fahrenheit when required.



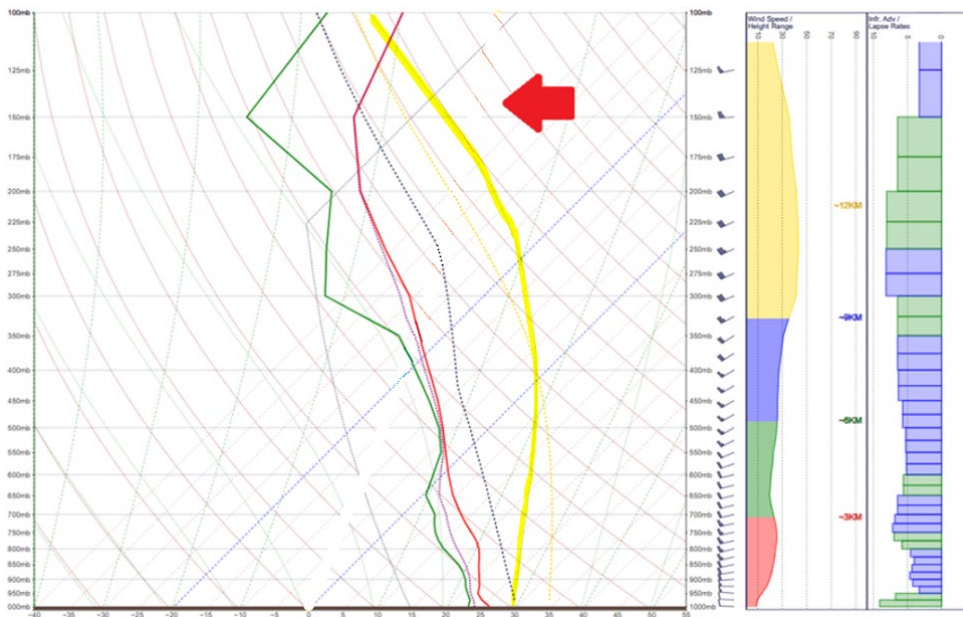
Dry Adiabats

Dry adiabats are lines depicting the pressure and temperature change rate for rising or sinking dry air. They are slightly curved, brown lines from the lower right to the upper left.



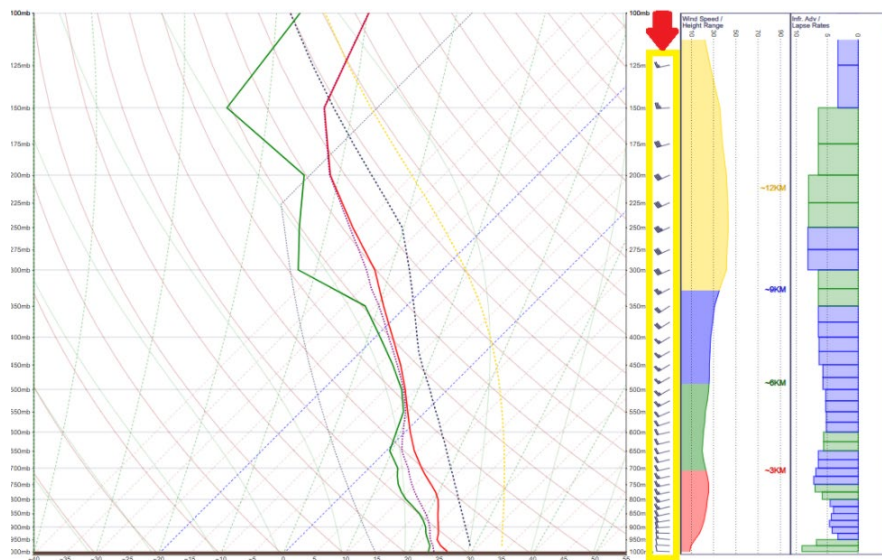
Saturation (Moist) Adiabats

Saturation adiabats are lines depicting the pressure and temperature change rate for rising or sinking moist air. They are slightly curved, solid green lines from the lower right to the upper left. Both saturation and dry adiabats parallel each other when the temperature is -55 degrees Celsius or colder.



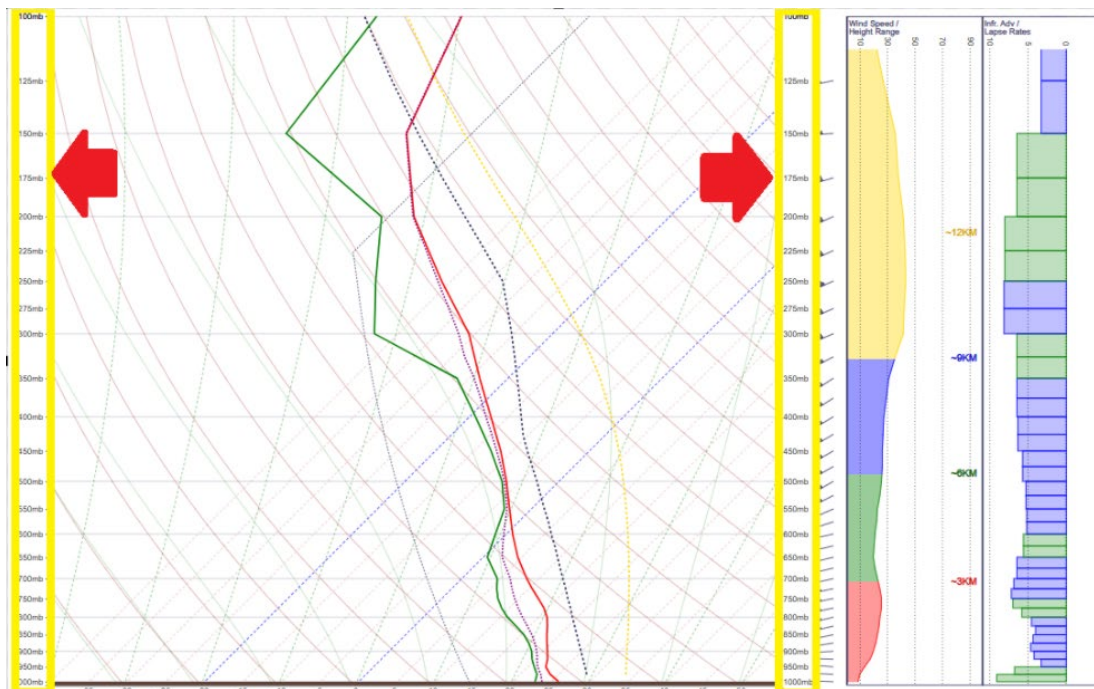
Wind Scales

The vertical staffs on the right side of the diagram depict wind speed and direction using wind barbs. Each wind barb may have a number plotted next to it. The number represents the tens unit for the wind direction.



Altitude Scale

On each side of the diagram, there will be an altitude scale measured in mb. Use the table data to determine the height in feet based on the mb height on the diagram.



Standard Atmosphere Heights

The standard height for each pressure surface is on the left side of the diagram. The heights are MSL with meters in brackets and feet in parenthesis. The standard heights in feet are:

- 850 mb – 5,000 feet
- 700 mb – 10,000 feet
- 500 mb – 18,000 feet
- 300 mb – 30,000 feet

Exercise / Practice: Skew-T Layout

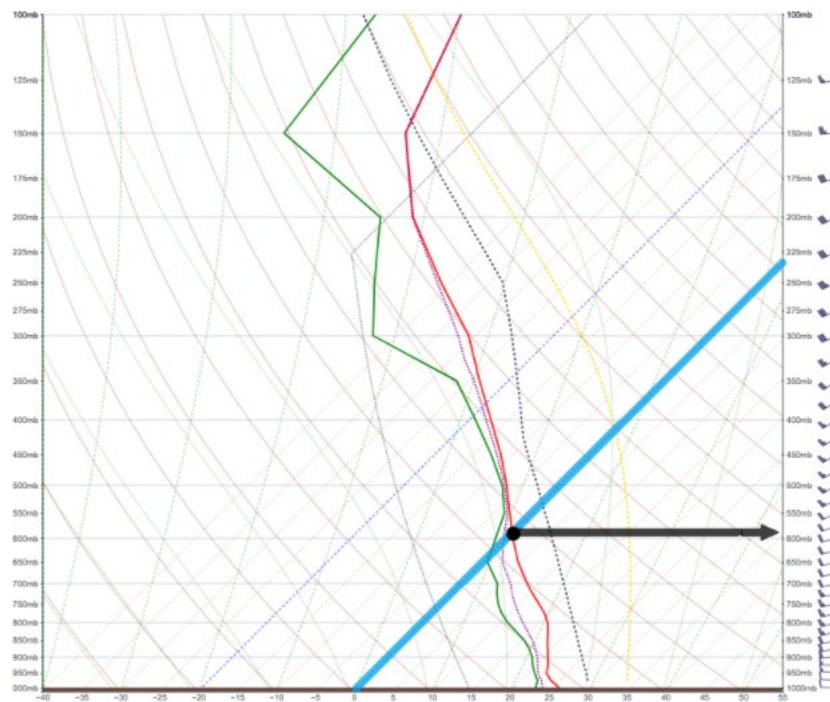
SKEW-T Analysis Techniques

Temperature & Dew Point Lines

There are two basic lines plotted on a Skew-T from which we can derive much information. These represent the dew point, which is calculated from the relative humidity (left line), and air temperature (right line).

Freezing Level

The freezing level is the height of the zero-degree isotherm. More specifically, it is the height where the temperature curve crosses the diagram's zero-degree isotherm. This may occur once or at multiple levels. Determine the freezing level by starting at the surface temperature. Follow the temperature curve upward until it crosses the zero-degree isotherm, then use the altitude scale to obtain the freezing level height to the nearest 100 feet.

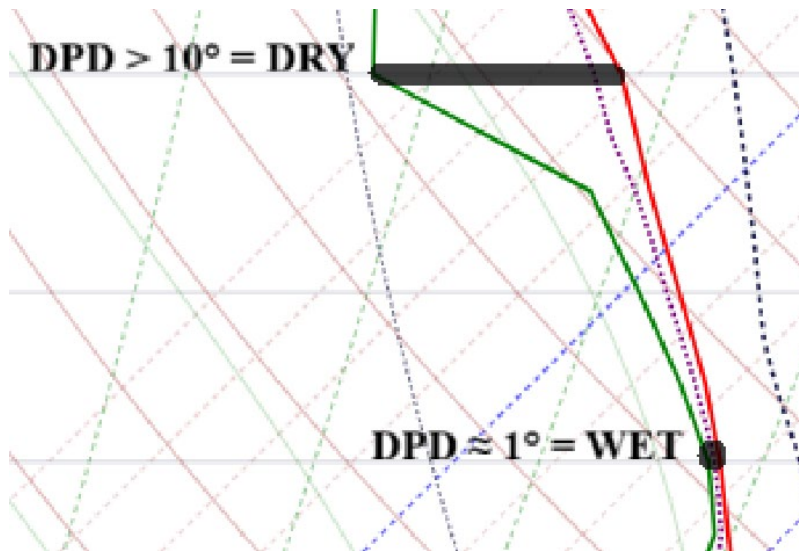


Available Moisture

You can identify the amount of available moisture at a location by using the areas dew-point depression (DPD). The distance between the temperature and dew-point curves outlines layers where there is enough moisture to fuel developing storms.

- 5+ (Dry)

- 3 to 4 (Moist)
- 1 to 2 (Wet)
- 0 (Fully Saturated)

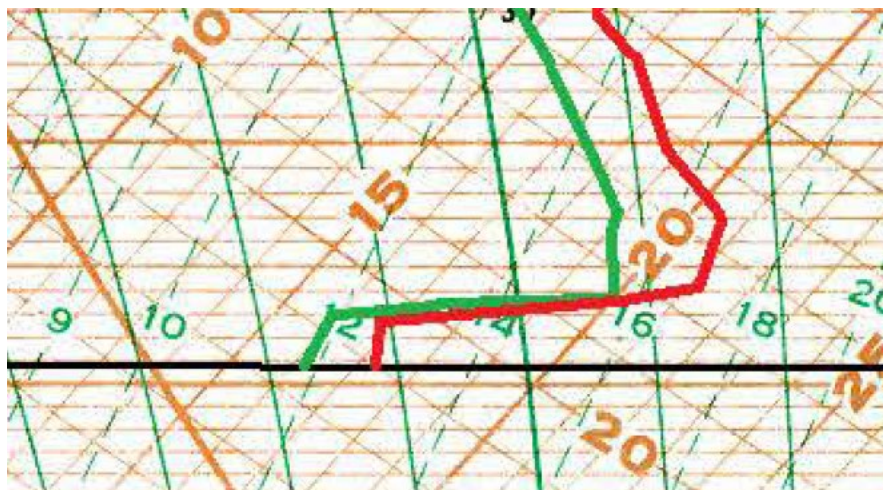


Exercise / Practice: Temp / DP / Moisture

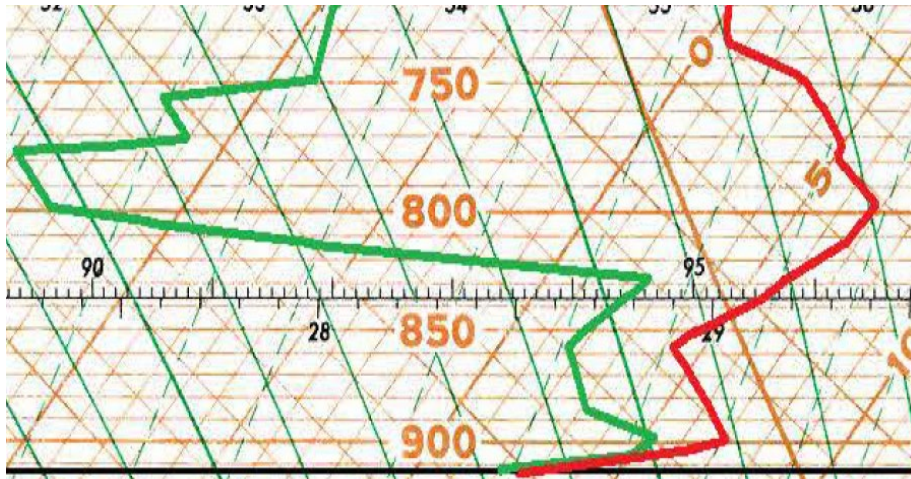
Inversions

In the troposphere, temperature will generally decrease (cool) with height. Inversions describe the areas of the atmosphere where the temperature increases (warms) with height. There are certain features that cause this to happen. There will not always be an inversion, but when an inversion is present, it is categorized as one of the following types.

Frontal: A frontal inversion is associated with surface fronts. The temperature and dew-point curves warm (increase) with height.

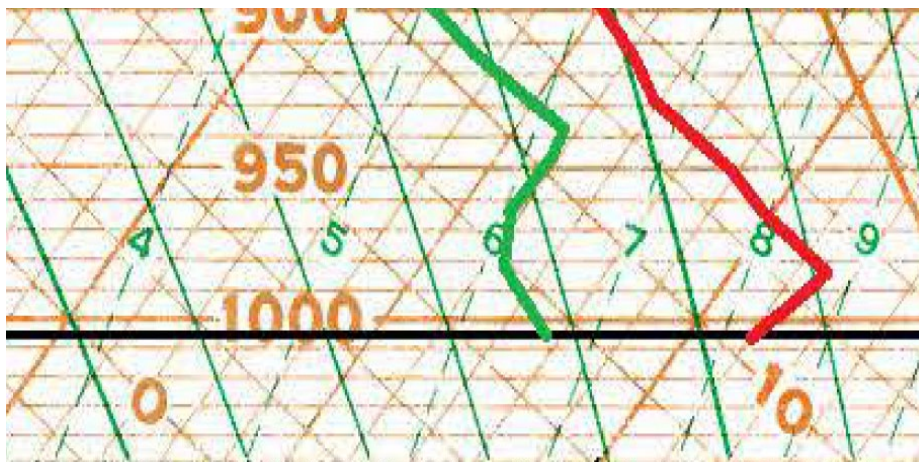


Subsidence: A subsidence inversion is associated with downward vertical motion. Sinking air adiabatically warms the troposphere creating the inversion. Dew-point depressions increase causing the temperature and dew-point curves move away from each other.



Radiation: Radiation inversions form due to nighttime cooling. The main difference between it and the others is radiation inversions are surface based meaning the base of the inversion is always at the Earth's surface. The orientation of the dew-point curve does not matter. It may increase in height looking like a frontal inversion, or it may decrease like a subsidence inversion.

- Near sunset the surface temperature begins to cool.
- Near midnight, the surface continues to cool as the earth radiates the heat absorbed during the day to create warmer temperatures above the surface and a radiation inversion develops.
- Before sunrise the surface temperature is at its coldest and the radiation inversion to reach its maximum intensity.
- After sunrise the surface begins to absorb heat. This causes the surface temperature to eventually become warmer than the atmosphere aloft. When the atmosphere is no longer warming with height the inversion has dissipated.
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Exercise / Practice: Inversions

Skew-T Indices

Convective Condensation level (CCL)

The CCL identifies the base of cumuliform clouds formed by free convection

Surface Data

Surface	26.16C	1018mb	33ft
SFC LCL	17.57C	931mb	2575ft
ML CCL	14.79C	836mb	5596ft

Lifted Condensation Level (LCL)

The LCL identifies the base of stratiform clouds formed by forced convection.

Surface Data

Surface	26.16C	1018mb	33ft
SFC LCL	17.57C	931mb	2575ft
ML CCL	14.79C	836mb	5596ft

Level of Free Convection (LFC)

The LFC is the height where air parcels are warmer than their environment. They can rise without any atmospheric assistance enhancing upward vertical motion.

Significant Levels

MU Level	25.07°C	1018mb	33ft
MU LCL	17.57°C	931mb	2575ft
MU LFC	17.57°C	931mb	2575ft
EQ Level	-20.77°C	390mb	25347ft
MP Level	-45.68°C	240mb	36546ft
TROP	-61.69°C	150mb	46422ft

Equilibrium Level (EL)

The EL is the top of the LFC. Air parcels cool to match their surrounding atmosphere and no longer rise on their own. It represents the top of a thunderstorm's updraft and often can be higher than the tropopause.

Significant Levels

MU Level	25.07°C	1018mb	33ft
MU LCL	17.57°C	931mb	2575ft
MU LFC	17.57°C	931mb	2575ft
EQ Level	-20.77°C	390mb	25347ft
MP Level	-45.68°C	240mb	36546ft
TROP	-61.69°C	150mb	46422ft

Convective Available Potential Energy (CAPE)

CAPE is the measurement of how unstable the atmosphere is. If you remember in the last unit, one of the ingredients needed for thunderstorm development was instability. The CAPE values on a SKEW-T can give you a tangible number to correlate how unstable you're the atmosphere is at a location. This is the area on a sounding between the temperature curve and the line connecting the LFC to the EL along the moist adiabat curve. The largest CAPE will occur in the warm sector of a mid-latitude cyclones, in the updraft region of a thunderstorm, an increase in low level warm air advection, daytime heating, or upper-level cold air advection. The things that cause atmospheric instability you learned previously will result in a higher CAPE value.

- 1 to 1,500 (Positive CAPE)
- 1,500 to 2,500 (Large CAPE)
- 2,500+ (Extreme CAPE)

Parcel Energy

SFC CAPE	97.12
SFC CINH	-6.04
ML CAPE	292.58
ML CINH	-3.7
MU CAPE	292.58
MU CINH	-3.7
DCAPE	614.47

Parcel Energy

SFC CAPE	1944.91
SFC CINH	0
ML CAPE	2080.16
ML CINH	-2.29
MU CAPE	2080.16
MU CINH	-2.29
DCAPE	642.35

Convective Inhibition (CINH)

CINH is the opposite of CAPE. CINH is a numerical value representing atmospheric stability, or how much resistance to upward vertical motion there is. A dry boundary layer will lead to high values of CINH and stability.

- Less than 50 (Low)
- 50-200 (Moderate)
- Above 200 (High)

Parcel Energy

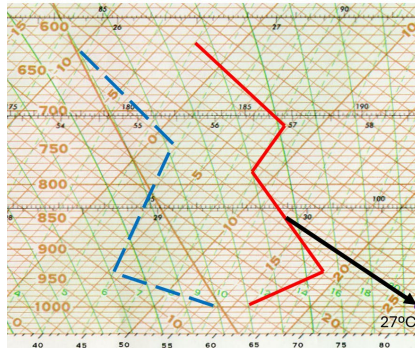
SFC CAPE	1346.63
SFC CINH	-0.25
ML CAPE	1491.75
ML CINH	-31.88
MU CAPE	1491.75
MU CINH	-31.88
DCAPE	518.06

Maximum Temperature

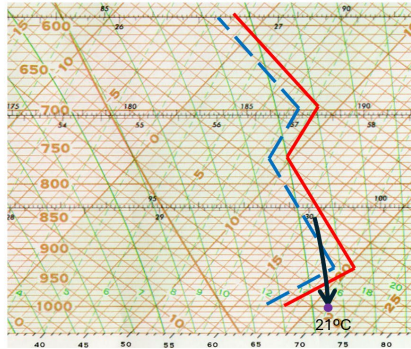
To determine the maximum temperature, you must use the cold Skew-T sounding. In the United States that is at 12Z. You also need to know the forecast sky cover during the day. When the forecast has broken to overcast skies, draw a line from the 850 mb temperature parallel to the

moist adiabats down to the surface. The temperature at that point is the maximum temperature in Celsius. If the forecast is predominately clear to scattered skies, draw a line from the 850 mb temperature parallel to the dry adiabats down to the surface. The temperature at that point is the maximum temperature in Celsius.

Clear-Scattered



Broken-Overcast



Convective Temperature

Convective temperature is the temperature required for cumuliform clouds to form. If your maximum temperature for the day reaches or exceeds the convective temperature, then daytime heating and free convection will happen resulting in potential thunderstorm development. The farther away your maximum temperature is from your convective temperature, the less likely free convection is to occur. In this case, only forced convection would produce enough lift for thunderstorm development.

Instability

CONVT	29.78
MU LI	-3.13
MU TTI	47.2
MU KI	[No Data]
SSI	0.09
MU THI	40.26
LSI	0

Showalter Stability Index (SSI)

This is the standard stability index for determining the potential for thunderstorm development.

- $> +3$ (Thunderstorms not likely)
- $+3$ to $+1$ (Thunderstorms Possible)
- 0 to -3 (Thunderstorms likely)
- -4 to -6 (Severe thunderstorms likely)

- < -6 (Tornadoes likely)

Instability

CONVT	29.81
MU LI	-2.64
MU TTI	46.16
MU KI	[No Data]
SSI	-0.01
MU THI	40.56
LSI	0

Total Totals (TT)

This index determines potential thunderstorm intensity.

- < 44 (convection not likely)
- 44 to 50 (convection likely)
- 51 - 52 (isolated severe storms)
- 53 - 56 (widely scattered severe storms)
- > 56 (scattered severe storms)

Instability

CONVT	29.92
MU LI	-2.57
MU TTI	46.07
MU KI	[No Data]
SSI	0.12
MU THI	40.53
LSI	0

K-Index (KI)

The KI is primarily applicable in the prediction of air mass thunderstorms. Low values of the K Index indicate low potential for air mass convection, while higher values indicate high potential.

- < 15 (Convection not likely)
- 15 to 25 (Small potential for convection)
- 26 to 39 (Moderate potential for convection)

- 40+ (High potential for convection)

Instability	
CONVT	30.12
MU LI	-2.22
MU TTI	44.31
MU KI	31.15
SSI	1.74
MU THI	36.82
LSI	0

Lifted Index (LI)

The LI is the temperature difference between the environmental temperature at 500mb and the air parcel temperature at 500mb. This calculation is another determination for the level of instability in the atmosphere.

- 0 or greater (stable)
- -1 to -4 (marginal instability)
- -5 to -7 (large instability)
- -8 to -10 (extreme instability)
- -11 or less (ridiculous instability)

Instability	
CONVT	27.51
MU LI	-3.58
MU TTI	48.46
MU KI	[No Data]
SSI	-0.97
MU THI	41.05
LSI	0

Hail & Wet-Bulb Zero (WBZ)

When predicting hail, the main factors to consider are the CAPE and the Wet-Bulb Zero height. The CAPE determines the potential size of hailstones before they fall. Higher CAPEs lead to hailstones being thrown higher vertical distances into the updraft.

The wet-bulb zero height is a function of how much dry air there is in the mid-levels of the atmosphere. On a Skew-T, the larger the dewpoint depression, the higher your wet-bulb zero height will be. Heights between 7,000 and 10,500 feet correlate well with large hail.

WBZ heights less than 10,500 feet will typically result in the Max Hail Index being the same or close in value to the Expected Hail value. If WBZ heights are above 10,500 feet, there will be a larger difference in those values due to the hail having more time to melt before reaching the surface.

Significant Height Levels

-20C Hgt	21909ft
FZ Level	10998ft
WBZ Hgt	9923ft

Significant Height Levels

-20C Hgt	21846ft
FZ Level	11129ft
WBZ Hgt	10717ft

Hail Indices

FAW-MIL HI	1
R&M HI	0
VIL-D HI	1
6KM SHEAR	22
SHiP	0
MAX HAIL	1
EXPCT HAIL	1

Hail Indices

FAW-MIL HI	1
R&M HI	0
VIL-D HI	1
6KM SHEAR	20
SHiP	0
MAX HAIL	1
EXPCT HAIL	0

Precipitable Water (PW)

Value of precipitable water in inches. This is the total amount of water vapor in a vertical column of the atmosphere.

- Greater than 1.75" represents a wet sounding.
- .75" to 1.75" represent a moist sounding.
- Less than 0.75" represents a dry sounding.

Precipitation

1-hr Precip	[No Data]
3hr Conv Precip	0.03
Prcp Water	2.14"

Thunderstorm Gust Speed

This shows gust speeds based on temperature differences and mean wind speeds. They can help forecasters assess the severity of potential wind gusts associated with thunderstorms. There are different methods designed to depict convective wind speeds and downbursts based on the type of thunderstorm present or expected. All methods are depicted in the skew-t indices in knots of potential gust speeds.

- T1 Gust – used for scattered or vicinity thunderstorms
- T2 Gust – used for intense squall lines and widespread thunderstorms
- Snyder Index – used for air-mass thunderstorms

CONV Winds (kt)	
T1 GUST	52
T2 GUST	38
WINDEX	30.61
Snyder	28
VILD Wnd	31.25
WMSI	9.79
Conv Gust	24.1

Exercise / Practice: Indices

Practice: Skew-T

PC: Skew-T